

Course - Bachelor of Arts (Economics) Honours

Course Subject - Introductory Econometrics

Semester – IV

Regression Analysis

Topic – How does gender-specific employment in agriculture influence agricultural productivity and its contribution to GDP across diverse countries?

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INTRODUCTION

The intricate relationship between gender-specific employment in agriculture and its consequential impact on agricultural productivity and economic growth has garnered significant attention in contemporary discourse. This regression analysis project delves into understanding the nuanced dynamics between gender-specific employment in agriculture and its influence on agricultural productivity, as well as its contribution to Gross Domestic Product (GDP) across diverse countries.

At the heart of this analysis lies the dependent variable, "Agriculture, value added (% of GDP)," which serves as a key indicator of the agricultural sector's contribution to the overall economic output of a nation. This variable encapsulates the significance of agriculture within national economies and provides a comprehensive measure of its relative importance.

To unravel the intricate web of factors influencing agricultural productivity and its contribution to GDP, this study examines two crucial explanatory variables: male employment in agriculture and female employment in agriculture. By dissecting the gender-specific aspects of agricultural labor, this analysis aims to uncover the differential impacts of male and female participation in the agricultural sector on overall productivity and economic performance.

Through rigorous regression analysis, this project seeks to illuminate the extent to which gender-specific employment patterns shape agricultural productivity and, consequently, influence the economic landscape of diverse countries. By elucidating these relationships, policymakers and stakeholders can gain invaluable insights into crafting targeted interventions aimed at fostering gender-inclusive agricultural development strategies, thereby unlocking the full potential of this vital sector in driving sustainable economic growth and development.

LITERATURE REVIEW

The paper examines how female participation in agriculture contributes to economic development in 33 selected African countries. This topic is important because women occupy a crucial position in the agricultural sector in Africa, contributing up to 40% of agricultural GDP despite their various other responsibilities like family and child-care. The study aims to align with the UN's Sustainable Development Goals 5 & 8 of promoting gender equality and decent work/economic growth. The authors review previous literature showing women's significant roles in agriculture globally, contributing over 40% of the world's food production. However, women often have fewer resources and assets like land compared to men. Past studies show mixed results on whether increasing women's participation in agriculture enhances economic development, possibly depending on access to education, training, and resources. The study uses panel data from 33 African countries from 2000-2018, obtained from the World Development Indicators and Human Development Index. The econometric model has economic development as the dependent variable, with key explanatory variables being female participation in agriculture, female population, and female education levels. Using pooled OLS and fixed effects regression methods, the authors find consistent results across the full sample and various regional sub-samples. Female participation in agriculture is statistically significant but negatively related to economic development across models. A 1% increase in female agricultural participation is associated with a 0.06% to 2.7% reduction in economic development, depending on the region. On the other hand, female education is significant and positively related to economic development across all models and regions. On average, a proportionate increase in female education is associated with around a 1.71% increase in economic development, holding other factors constant. The results imply that simply increasing women's participation in agriculture without improving their education, training and access to resources may actually hinder economic development in Africa. The negative relationship for female agricultural participation contrasts with prior expectations but aligns with some previous studies showing gender inequalities limiting women's productive potential. The authors conclude that while women play crucial roles in African

agriculture, increasing agricultural participation alone is insufficient for driving economic development. Investing in female education and human capital development are critical to realizing the productive benefits of higher female labor force engagement. Upgrading technology, management capabilities and support services in agriculture requires improving the skills of agricultural human resources. Some limitations are that the study did not account for potential endogeneity due to data constraints. The authors recommend future research using methods like GMM to address this. Overall, the study highlights the importance of closing gender gaps in education and access to productive resources, in addition to increasing female employment, for African economies to fully capitalize on women's agricultural contributions.

FINDINGS:

The relationship between female employment in agriculture and its contribution to GDP is examined in the paper. Despite women contributing up to 40% of agricultural GDP in Africa, the study finds a negative relationship between female participation in agriculture and economic development. However, female education is positively correlated with economic development, suggesting that improving education and access to resources for women is crucial for maximizing their productive contributions to the economy.

Keywords: Agriculture, SGD 5 & 8, African women, Africa

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The paper examines the role of male and female employment in agriculture and its impact on agricultural productivity in Nigeria. Agriculture is an important sector in Nigeria, contributing around 21% of GDP in 2018. However, the sector has been underperforming, with 21.2% of people in Africa estimated to be undernourished between 2011-2013. The contribution of women to agricultural labor in Africa is often quoted as 60-80%. However, in Nigeria the contribution of females in agriculture is substantially lower at 37%, with males comprising 63% of agricultural employment. There has been a consistent decreasing trend in female employment in Nigerian agriculture compared to males over recent decades. The paper reviews literature showing that gender gaps and inequalities in areas like access to resources, technology adoption, and economic opportunities negatively impact agricultural productivity in Africa. Closing these gender gaps through gender-responsive policies and programs is seen as key to increasing productivity, food security and poverty reduction. Several studies cited find evidence of gender productivity gaps in Nigerian and other African agriculture, with male farmers outperforming females due to factors like more experience, better access to labor, land and inputs like fertilizer. However, other studies show higher adoption of technologies like improved seeds among female farmers. The paper uses an Autoregressive Distributed Lag (ARDL) model and time series data from 1990-2019 to analyze the impact of male and female agricultural employment, along with other factors like trade, inflation, commercial bank lending to agriculture, and capital formation on total agricultural productivity in Nigeria. The ARDL results indicate a long-run cointegrating relationship between the variables. In the long-run, commercial bank credit to agriculture, capital formation, agricultural machinery and arable land have a positive relationship with agricultural productivity, as expected. The error correction results suggest around 70% of any disequilibrium in the model is corrected in the next period. Differenced terms for agricultural credit schemes, total agricultural employment, and lagged agricultural machinery are also significant determinants of productivity in the short-run. The study finds that increasing employment opportunities and empowerment for women in agriculture, along with access to credit, machinery and arable land

can boost overall agricultural productivity in Nigeria. It recommends policies promoting gender equality and female participation in agriculture to improve economic growth and development. The diagnostic tests indicate the ARDL model is stable and robust. The study concludes that while female contribution to agricultural labor is lower than males in Nigeria, enhancing opportunities for women through gender-focused policies can significantly increase productivity and development in the agricultural sector.

FINDINGS: The relationship between female employment and its contribution to GDP in Nigeria's agriculture sector is that although females contribute less than males to agricultural labor, enhancing opportunities for women through gender-focused policies can significantly increase productivity and development, thereby positively impacting GDP.

LITERATURE GAP

The existing literature provides insights into the impact of gender-specific agricultural employment on economic development, particularly in African countries like Nigeria and across the continent. However, there is a notable gap in understanding how this relationship varies across diverse nations globally. While studies have focused on specific countries or regions, there is a lack of comprehensive analysis that encompasses a wide range of countries with varying socio-economic contexts. Additionally, existing research predominantly examines the role of female participation in agriculture, leaving a gap in understanding the comparative contributions of male and female employment to agricultural value-added across different nations. Addressing this gap would provide valuable insights into the nuanced dynamics of gender-specific agricultural employment and its implications for GDP growth in diverse national contexts, thereby informing more targeted and effective policy interventions aimed at promoting gender equality and enhancing agricultural productivity worldwide.

DATA

Country	Employment in agriculture, male (% of male employment)	Employment in agriculture, female (% of female employment)	Agriculture, value added (% of GDP)
Afghanistan	41.6	58.8	76
Albania	32.4	41.6	50.8
Algeria	11.1	3.7	-3.7
Angola	47.3	59.6	13.6
Argentina	10.4	3.2	6.6
Armenia	36.5	61.3	10.4
Australia	3.3	1.7	2.7
Austria	4.1	3.2	1.4
Azerbaijan	30.8	41.3	4.8
Bahamas, The	5.1	1.1	0.6
Bahrain	1.2	0	0.3
Bangladesh	29.6	57.1	11.2
Barbados	4.7	1	1.5
Belarus	14.6	7.6	7.7
Belgium	1.2	0.6	0.6
Belize	22.6	6.8	7.8
Benin	41.6	15	26.9
Bhutan	42.9	62.4	15.8
Bolivia	27.3	27.6	12.5
Bosnia and Herzegovina	16.4	20.5	4.8
Botswana	20.2	8.7	1.8
Brazil	12.5	4	6.8
Brunei Darussalam	2.8	0.7	1.1
Bulgaria	8.6	4.3	3.9
Burkina Faso	72.8	76.9	18.5
Burundi	77.9	93	27.6
Cabo Verde	14.8	5.5	4.8
Cambodia	39.1	40.7	22.2
Cameroon	38.5	48.3	17
Canada	2	1	1.8
Central African Republic	66.6	77	36.2
Chad	69.7	70.1	22.6
Chile	12.1	4.8	3.5
Colombia	22.7	6.3	8.3
Comoros	36.3	35.6	36.4
Congo, Dem. Rep.	48.9	65.3	17.4
Congo, Rep.	32.9	32.4	8
Costa Rica	19.4	9	4.1
Cote d'Ivoire	50.5	39.8	16.7
Croatia	7.7	4.4	3.1
Cuba	23.9	7	0.8
Cyprus	3.3	1.3	1.6

Czechia	3.4	1.7	1.9
Denmark	3.2	1.1	1.2
Djibouti	1.3	0.9	1.8
Dominican Republic	13.5	1.4	5.8
Ecuador	31.5	27.2	8.8
Egypt, Arab Rep.	21.1	21.4	10.9
El Salvador	25.3	3.4	4.7
Equatorial Guinea	53	59.2	2.7
Estonia	59.2	70.1	2.5
Eswatini	4.7	1.5	8.6
Ethiopia	71.9	57.5	37.6
Fiji	34.1	19.7	12.6
Finland	5.4	2.1	2.3
France	3.5	1.5	1.9
Gabon	23.5	42	5.6
Gambia, The	34.9	63.9	22.6
Georgia	36.3	40.2	6
Germany	1.5	0.8	0.9
Ghana	42.8	37.3	19.6
Greece	11.9	10.1	3.8
Guatemala	40.4	11.1	9.3
Guinea	58.4	64.7	27.3
Guinea-Bissau	47.6	55.3	30.9
Guyana	20.5	7.3	10
Haiti	59	31.1	20.3
Honduras	41.2	12.8	12.6
Hungary	6.3	2.8	3.2
Iceland	6.3	1.4	4.3
India	36.2	53.9	16.7
Indonesia	30.1	26.6	12.4
Iran, Islamic Rep.	16.4	19.8	12.8
Iraq	7.7	14.7	2.9
Ireland	7.2	1.2	1.1
Israel	1.3	0.5	1.3
Italy	5	2.4	1.8
Jamaica	20.9	8.2	8.1
Japan	3.6	2.8	1
Jordan	3.8	1.1	4.8
Kazakhstan	15.2	12.1	5.2
Kenya	31.5	35.8	21.2
Korea, Rep.	5.4	4.8	1.6
Kuwait	2.6	0.1	0.5
Kyrgyz Republic	19	31.5	10.9
Lao PDR	63.3	65.6	14.9
Latvia	10.3	4.3	5
Lebanon	4.4	1.5	1.4
Lesotho	38.7	18.6	6.2
Liberia	37.6	42.9	36.2
Libya	11.3	5.8	1.6

Lithuania	8.8	4.1	4
Luxembourg	0.9	0.4	0.2
Madagascar	73	70.8	21.9
Malawi	58	67.9	21.8
Malaysia	12.8	6.1	8.9
Maldives	9.9	2.3	4.9
Mali	66.1	64.9	36.4
Malta	1.4	0.4	0.9
Mauritania	35.5	30.8	22.2
Mauritius	6.7	3.8	3.6
Mexico	18.5	4.3	4
Moldova	57.4	58.9	8.3
Mongolia	29	26.3	13
Montenegro	7.6	6.6	6
Morocco	26.8	50.6	10.3
Mozambique	60.5	80.4	26.7
Myanmar	53.7	43.5	22.3
Namibia	23.6	20.5	8.4
Nepal	53.8	76.3	21.1
Netherlands	2.7	1.4	1.5
New Zealand	7.7	3.7	5.7
Nicaragua	40.8	8.2	16.8
Niger	72.9	69.3	42
Nigeria	45.4	30	23.7
Norway	2.9	1	1.7
Oman	5.4	0.2	1.8
Pakistan	29.5	69.2	22.4
Panama	18.6	8.3	2.6
Papua New Guinea	24.3	14.2	16.9
Paraguay	22.7	13	11
Peru	27	23.8	7.8
Philippines	28.7	13.6	9.5
Poland	10	8.1	2.8
Portugal	7.3	3.6	1.9
Puerto Rico	2.1	0.1	0.7
Qatar	1.8	0	0.3
Romania	21.4	21	4.5
Russian Federation	7.6	4	3.9
Rwanda	38.8	57.3	24.9
Samoa	27.4	9.1	10
Sao Tome and Principe	20.6	11.7	13.9
Saudi Arabia	3.7	2	2.4
Senegal	26.1	18	15.5
Serbia	17.5	13.3	6.5
Sierra Leone	45.8	43.8	60.4
Singapore	0.1	0.1	0
Slovak Republic	4	1.3	2.2
Slovenia	5	3.5	1.8
Solomon Islands	37.2	38.9	34.1

South Africa	15.5	17.4	2.8
Spain	5.7	2	2.3
Sri Lanka	24.9	26.2	8.7
St. Lucia	16.1	3.1	1.4
St. Vincent and the Grenadines	14.5	4.1	4.7
Sudan	36.6	53.4	5
Suriname	8.8	6	8.1
Sweden	2.6	0.7	1.5
Switzerland	3.1	1.9	0.6
Syrian Arab Republic	17	10.7	27.8
Tajikistan	34.2	62.1	22.4
Tanzania	64.1	67.6	24.3
Thailand	34.1	28.3	8.8
Timor-Leste	42.3	46.3	10.2
Togo	37	26.4	18.3
Tonga	42.3	12	16.3
Trinidad and Tobago	4.3	1.4	1.1
Tunisia	16	9.3	9.8
Türkiye	14.9	25.1	6.5
Turkmenistan	23.7	23.2	11.7
Uganda	63.3	74	24
Ukraine	17.3	12.9	8.2
United Arab Emirates	2.7	0.1	0.8
United Kingdom	1.5	0.6	0.8
United States	2.3	1	1
Uruguay	12.1	3.7	7.3
Uzbekistan	27.5	23.8	23.5

Data source : World bank

ECONOMETRIC MODEL

The econometric model used in this study is a multi linear regression model based on Classical Linear Regression Model (CLRM) assumptions. The estimators have been obtained using the Ordinary Least Squares Method (OLS).

The assumptions of the CLRM are :

- The regression model is linear in parameters. It may or may not be linear in the variables.
- The explanatory variable is uncorrelated with the disturbance term.
- The conditional mean of disturbance term with respect to the explanatory variable is zero.
- The variance of each u_i is constant, or homoscedastic.
- There is no correlation between two error terms, that is, there is no autocorrelation.
- There is no perfect multicollinearity.
- The regression model is correctly specified, i.e, there is no specification biasedness.
- All explanatory variables are not stochastic.

VARIABLES

Dependent Variable

➤ **Agriculture, value added (% of GDP):**

This variable represents the contribution of the agricultural sector to the Gross Domestic Product (GDP) of a country. It measures the value added by agriculture, forestry, and fisheries to the overall economic output of a nation as a percentage of its total GDP.

Explanatory Variables

➤ **Employment in agriculture, male (% of male employment):**

This variable represents the percentage of male workers who are employed in the agricultural sector. It indicates the proportion of men engaged in agricultural activities compared to all employed males.

➤ **Employment in agriculture, female (% of female employment):**

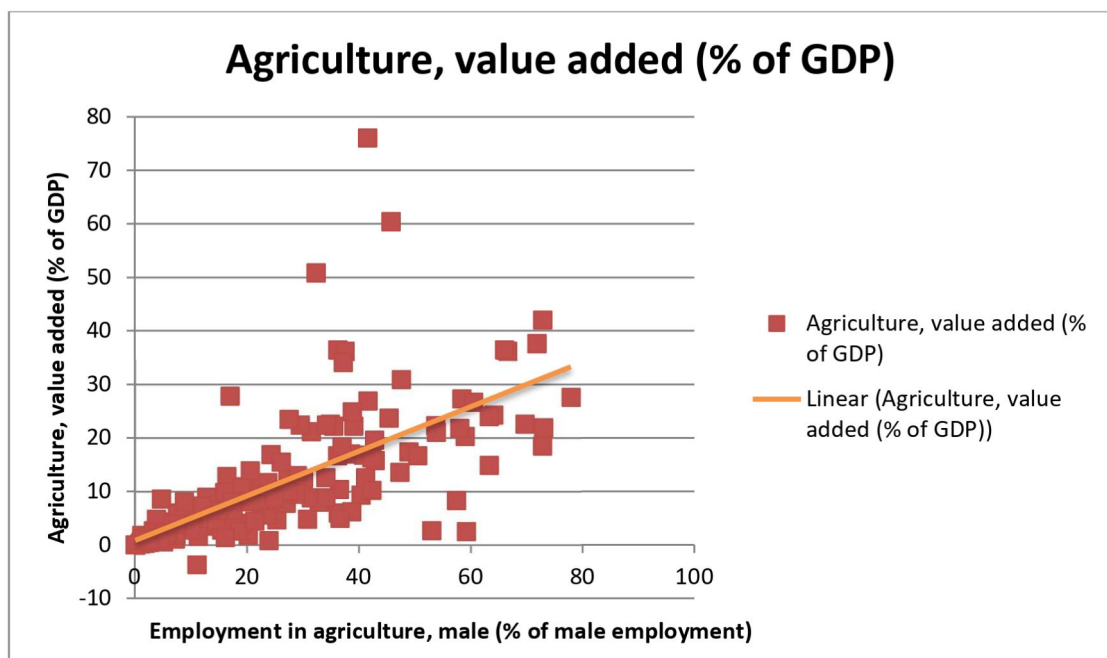
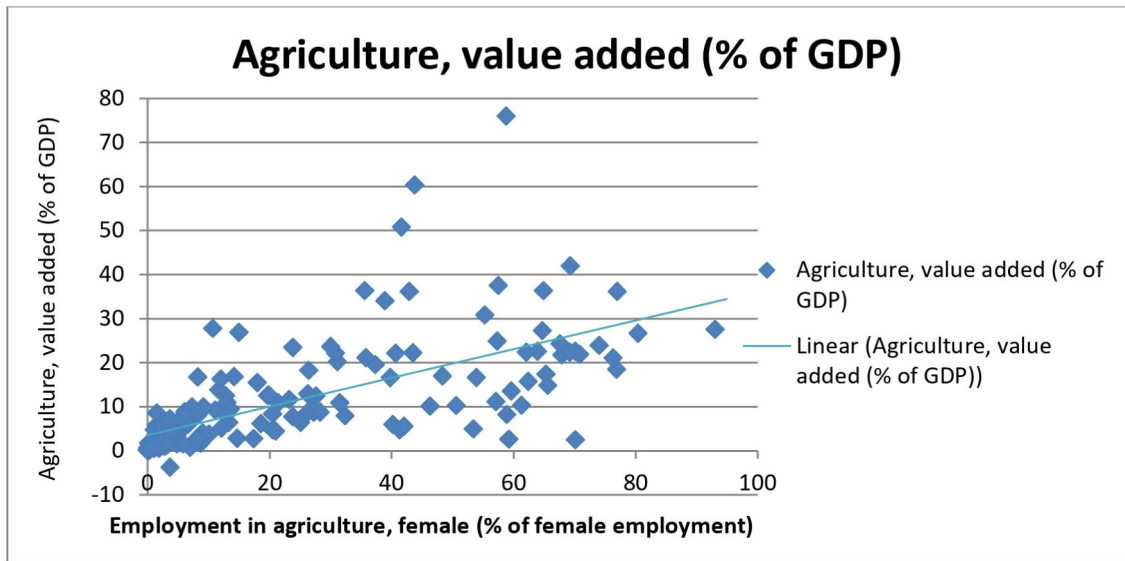
This variable represents the percentage of female workers who are employed in the agricultural sector. It indicates the proportion of women engaged in agricultural activities compared to all employed females.

Expected Signs

In this regression analysis, I anticipate a positive sign for both coefficients, B_1 and B_2 . A positive coefficient for male employment in agriculture (B_1) suggests that an increase in male participation in the agricultural workforce is likely to positively influence agricultural productivity and its contribution to GDP. Similarly, a positive coefficient for female employment in agriculture (B_2) indicates that a higher proportion of female labor in agriculture is expected to contribute positively to agricultural productivity and GDP. These expected positive signs align with the notion that diverse gender participation in agriculture can enhance overall productivity and economic output.

We have chosen a simple multi linear model based on the presence of linear relationship between the variables plotted on scatter plots

SCATTER PLOT



MODEL

The equation of the model can be formulated as

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

where,

- Y_i is the dependent variable,
- X_1 is the independent variable,
- β_1 is the intercept term,
- β_2 is the slope coefficient of independent variable 1, and
- β_3 is the slope coefficient of independent variable 2.

$$Gdp_{agri} = \beta_0 + \beta_1 * emp_m + \beta_2 * emp_f + \varepsilon$$

where:

- Gdp_{agri} represents Agriculture, value added (% of GDP).
- emp_m represents Employment in agriculture, male (% of male employment).
- emp_f represents Employment in agriculture, female (% of female employment).
- ε is the error term.

Summary Statistics

<i>Employment in agriculture, male (% of male employment)</i>		<i>Employment in agriculture, female (% of female employment)</i>	
Mean	24.15868263	Mean	22.56167665
Standard Error	1.547936213	Standard Error	1.896213508
Median	20.2	Median	11.1
Mode	7.7	Mode	1
Standard Deviation	20.00374436	Standard Deviation	24.5044789
Sample Variance	400.1497886	Sample Variance	600.4694863
Kurtosis	-0.245201624	Kurtosis	-0.362215231
Skewness	0.800120048	Skewness	0.974017429
Range	77.8	Range	93
Minimum	0.1	Minimum	0
Maximum	77.9	Maximum	93
Sum	4034.5	Sum	3767.8
Count	167	Count	167

<i>Regression Statistics</i>	
Multiple R	0.761574843
R Square	0.579996241
Adjusted R Square	0.574874244
Standard Error	6.840064914
Observations	167

	<i>Coefficients</i>	<i>Standard Error</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	1.23337822	1.065174551	0.248583526	0.869845761	3.336602201	0.869845761	3.336602201
Male Employment in agriculture	0.297025176	0.073568095	8.27815E-05	0.151762429	0.442287922	0.151762429	0.442287922
Female Employment in agriculture	0.109209025	0.060055852	0.070818677	0.009373331	0.227791381	0.009373331	0.227791381

ANOVA				
	<i>df</i>	<i>SS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	11760.23438	80.35507702	4.72644E-25
Residual	164	12000.97436		
Total	166	23761.20874		

REGRESSION EQUATION

The equation obtained by regressing Gdp_{agri} on emp_m and emp_f is –

$$GDP_{agri}=1.233+0.297\times emp_m+0.109\times emp_f$$

Interpreting the coefficients:

- The intercept term (1.233) represents the estimated Agriculture value added (% of GDP) when both male and female employment in agriculture are zero. However, in practical terms, this value may not hold much meaning as it's unlikely to have zero employment in agriculture in a real-world scenario.
- The coefficient 0.297 for emp_m means that for each additional percentage point increase in male employment in agriculture, we expect the Agriculture value added (% of GDP) to increase by 0.297 percentage points, holding other variables constant.
- Similarly, the coefficient 0.109 for emp_f means that for each additional percentage point increase in female employment in agriculture, we expect the Agriculture value added (% of GDP) to increase by 0.109 percentage points, holding other variables constant.

Interpreting R²

R² value (0.580) indicates that approximately 58% of the variation in Agriculture value added (% of GDP) can be explained by the linear relationship with the percentage of male and female employment in agriculture. In other words, the model explains 58% of the variability in Agriculture value added (% of GDP) based on the variation in male and female employment percentages.

TESTING FOR SIGNIFICANCE OF SLOPE COEFFICIENTS

Null Hypothesis (H_0): There is no significant relationship between the percentage of male employment in agriculture and Agriculture value added (% of GDP), as well as between the percentage of female employment in agriculture and Agriculture value added (% of GDP).

Alternative Hypothesis (H_1): There is a significant relationship between at least one of the independent variables (percentage of male employment or percentage of female employment in agriculture) and Agriculture value added (% of GDP).

Now, let's conduct the hypothesis tests step by step for each independent variable:

1. Percentage of Male Employment (emp_m):

Null Hypothesis (H_0): $\beta_1 = 0$

Alternative Hypothesis (H_1): $\beta_1 \neq 0$

P-value: $p < 0.05$

Since the p-value associated with emp_m is less than 0.05 (assuming a significance level of 0.05), we reject the null hypothesis.

Therefore, there is sufficient evidence to suggest that there is a significant relationship between the percentage of male employment in agriculture and Agriculture value added (% of GDP).

Percentage of Female Employment

Null Hypothesis (H_0): $\beta_2 = 0$

Alternative Hypothesis (H_1): $\beta_2 \neq 0$

Since the p-value associated with $\text{emp}_f = 0.070818677$ is greater than 0.05 (assuming a significance level of 0.05), we fail to reject the null hypothesis. Therefore, there is not sufficient evidence to suggest that there is a significant relationship between the percentage of female employment in agriculture and Agriculture value added (% of GDP) at the 0.05 significance level.

While the statistical test did not yield significance, theoretical insights and broader datasets could reveal the substantial impact of female employment on agricultural GDP. Factors such as women's roles in agricultural labor, access to resources, and their contribution to agricultural productivity might not be fully captured in the current model. A more comprehensive analysis incorporating a wider range of variables and a larger dataset could elucidate the significant influence of female employment on agricultural GDP, aligning with existing theories and empirical evidence.

TESTING FOR JOINT SIGNIFICANCE OF VARIABLE

TESTING FOR OVERALL GOODNESS OF FIT

Null Hypothesis (H_0): $\beta_1 = \beta_2 = 0$

(where β_1 and β_2 are the coefficients for "Employment in agriculture, male (% of male employment)" and "Employment in agriculture, female (% of female employment)" respectively)

Alternative Hypothesis (H1): At least one of β_1 or β_2 is not equal to zero.

The F-test statistic is calculated as the ratio of the explained variance to the unexplained variance:

$$F = MSR / MSE$$

Where:

- MSR (Mean Square Regression) is the sum of squares due to regression,
- MSE (Mean Square Error) is the sum of squares of residuals

To conduct the F-test for joint hypothesis, we compare the calculated F-test statistic with the critical F-value at a chosen significance level and degrees of freedom.

Given:

- $MSR = 5880.11719$
- $MSE = 73.17667294$
- $df_{reg} = 2$ (degrees of freedom for regression)
- $df_{res} = 164$ (degrees of freedom for residuals)

$$F = 5880.11719 / 73.17667294$$

$$F \approx 80.36$$

Now, we need to find the critical F-value from the F-distribution table at a chosen significance level (usually 0.05) with degrees of

freedom for regression (2) and degrees of freedom for residuals (164).

Let's assume we chose a significance level of 0.05. From the F-distribution table, we find the critical F-value to be approximately 3.05.

Since the calculated F-value (80.36) is much greater than the critical F-value (3.05), we reject the null hypothesis.

This indicates that at least one of the independent variables, "Employment in agriculture, male (% of male employment)" or "Employment in agriculture, female (% of female employment)", has a significant effect on the dependent variable at the chosen significance level.

MULTICOLINEARITY

	<i>Male</i>	<i>Female</i>	Agriculture contribution in gdp
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Male	1		
Female	0.892442612	1	
Agriculture contribution in gdp	0.759449527	0.703416534	1

Correlation between explanatory variables= 0.892443

Variance inflation factor= 4.91290 664999

To address multicollinearity, you can consider the following measures:

- Remove one of the correlated variables: If two or more variables are highly correlated, you can remove one of them from the model. However, this should be done cautiously, ensuring that the variable being removed is not essential for the analysis or interpretation.
- Combine correlated variables: Sometimes, it's possible to create a new variable combines the information from the correlated variables. For example, if you have two variables measuring similar aspects of the same phenomenon, you could composite variable that captures both aspects.
- Collect more data: Increasing the sample size can sometimes help reduce multicollinearity by providing more information for the model to differentiate between correlated variables.
- Regularization techniques: Techniques like Ridge regression or Lasso regression can help mitigate multicollinearity by penalizing large coefficients, effectively their impact on the model.
- Principal Component Analysis (PCA): PCA is a dimensionality reduction technique used to transform correlated variables into a set of linearly uncorrelated variables called principal components. These components can then be used as predictors in regression model.

HETROSCEDASTICITY

<i>Country</i>	<i>Predicted Y</i>	<i>Residuals</i>
Afghanistan	20.01111622	55.98888378
Albania	15.40008937	35.39991063
Algeria	4.934431066	-8.634431066
Angola	21.79152694	-8.191526941
Argentina	4.67190893	1.92809107
Armenia	18.76931038	-8.369310384
Australia	2.399216643	0.300783357
Austria	2.800650322	-1.400650322
Azerbaijan	14.89208638	-10.09208638
Bahamas, The	2.868336545	-2.268336545
Bahrain	1.589808431	-1.289808431
Bangladesh	16.26115876	-5.061158765
Barbados	2.738605572	-1.238605572
Belarus	6.39993438	1.30006562
Belgium	1.655333846	-1.055333846
Belize	8.688768567	-0.888768567
Benin	15.22776092	11.67223908
Bhutan	20.79040144	-4.990401438
Bolivia	12.35633462	0.143665383
Bosnia and Herzegovina	8.343376121	-3.543376121
Botswana	8.183405292	-6.383405292
Brazil	5.38302902	1.41697098
Brunei Darussalam	2.14149503	-1.04149503
Bulgaria	4.257393541	-0.357393541
Burkina Faso	31.25498506	-12.75498506
Burundi	34.52807876	-6.928078764
Cabo Verde	6.230000462	-1.430000462
Cambodia	17.29186992	4.908130077
Cameroon	17.94364341	-0.943643408
Canada	1.936637597	-0.136637597
Central African Republic	29.42434987	6.775650126
Chad	29.59158565	-6.991585645
Chile	5.351586169	-1.851586169
Colombia	8.663866572	-0.363866572
Comoros	15.9032334	20.4967666
Congo, Dem. Rep.	22.88925867	-5.489258666
Congo, Rep.	14.54387892	-6.543878923
Costa Rica	7.978547859	-3.878547859
Cote d'Ivoire	20.57966881	-3.879668805
Croatia	4.000991785	-0.900991785
Cuba	9.0967431	-8.2967431
Cyprus	2.355533033	-0.755533033
Czechia	2.428919161	-0.528919161
Denmark	2.303988711	-1.103988711
Djibouti	1.717799072	0.082200928
Dominican Republic	5.39611073	0.40388927
Ecuador	13.56015675	-4.760156746

Egypt, Arab Rep.	9.83768257	1.06231743
El Salvador	9.119425856	-4.419425856
Equatorial Guinea	23.44088683	-20.74088683
Estonia	26.4728213	-23.9728213
Eswatini	2.793210085	5.806789915
Ethiopia	28.86900731	8.730992685
Fiji	13.51335451	-0.913354514
Finland	3.066653123	-0.766653123
France	2.436779874	-0.536779874
Gabon	12.80024891	-7.200248911
Gambia, The	18.57801357	4.021986432
Georgia	16.40559492	-10.40559492
Germany	1.766283204	-0.866283204
Ghana	18.01955239	1.580447612
Greece	5.870988968	-2.070988968
Guatemala	14.44541551	-5.145415506
Guinea	25.64547242	1.654527578
Guinea-Bissau	21.41103569	9.488964315
Guyana	8.11962021	1.88037979
Haiti	22.15426428	-1.854264281
Honduras	14.86869099	-2.268690989
Hungary	3.410422099	-0.210422099
Iceland	3.257529464	1.042470536
India	17.87205604	-1.172056045
Indonesia	13.07879608	-0.678796084
Iran, Islamic Rep.	8.266929803	4.533070197
Iraq	5.125844745	-2.225844745
Ireland	3.503010317	-2.403010317
Israel	1.674115461	-0.374115461
Italy	2.98060576	-1.18060576
Jamaica	8.336718403	-0.236718403
Japan	2.608454124	-1.608454124
Jordan	2.482203816	2.317796184
Kazakhstan	7.069590098	-1.869590098
Kenya	14.49935436	6.700645638
Korea, Rep.	3.361517491	-1.761517491
Kuwait	2.01656458	-1.51656458
Kyrgyz Republic	10.31694086	0.583059145
Lao PDR	27.19918391	-12.29918391
Latvia	4.76233634	0.23766366
Lebanon	2.704102532	-1.304102532
Lesotho	14.7595404	-8.559540396
Liberia	17.08659201	19.11340799
Libya	5.223175054	-3.623175054
Lithuania	4.294956771	-0.294956771
Luxembourg	1.544384489	-1.344384489
Madagascar	30.64821504	-8.748215043
Malawi	25.87613123	-4.076131232
Malaysia	5.701475525	3.198524475

Maldives	4.425108219	0.474891781
Mali	27.95440808	8.445591919
Malta	1.692897077	-0.792897077
Mauritania	15.14140994	7.05859006
Mauritius	3.638441194	-0.038441194
Mexico	7.197942782	-3.197942782
Moldova	24.7150349	-16.4150349
Mongolia	12.71930568	0.280694317
Montenegro	4.211549123	1.788450877
Morocco	14.71962961	-4.419629608
Mozambique	27.98380699	-1.283806987
Myanmar	21.93422276	0.365777239
Namibia	10.48195739	-2.081957388
Nepal	25.5459813	-4.445981305
Netherlands	2.18823883	-0.68823883
New Zealand	3.924545468	1.775454532
Nicaragua	14.2475194	2.552480597
Niger	30.45469899	11.54530101
Nigeria	17.99459196	5.705408039
Norway	2.203960255	-0.503960255
Oman	2.859155975	-1.059155975
Pakistan	17.55288545	4.847114548
Panama	7.664481401	-5.064481401
Papua New Guinea	10.00185815	6.898141848
Paraguay	9.39556704	1.60443296
Peru	11.85223277	-4.052232769
Philippines	11.24324351	-1.743243511
Poland	5.088223083	-2.288223083
Portugal	3.794814495	-1.894814495
Puerto Rico	1.868051992	-1.168051992
Qatar	1.768023537	-1.468023537
Romania	9.883106513	-5.383106513
Russian Federation	3.927605658	-0.027605658
Rwanda	19.01563219	5.884367812
Samoa	10.36567017	-0.365670169
Sao Tome and Principe	8.629842438	5.270157562
Saudi Arabia	2.550789421	-0.150789421
Senegal	10.95149776	4.548502236
Serbia	7.883798833	-1.383798833
Sierra Leone	19.62048658	40.77951342
Singapore	1.27400164	-1.27400164
Slovak Republic	2.563450657	-0.363450657
Slovenia	3.100735688	-1.300735688
Solomon Islands	16.53094584	17.56905416
South Africa	7.737505485	-4.937505485
Spain	3.144839773	-0.844839773
Sri Lanka	11.49058156	-2.79058156
St. Lucia	6.35403153	-4.95403153
St. Vincent and the	5.988000274	-1.288000274

Grenadines		
Sudan	17.9362616	-12.9362616
Suriname	4.502453919	3.597546081
Sweden	2.082089995	-0.582089995
Switzerland	2.361653413	-1.761653413
Syrian Arab Republic	7.45134278	20.34865722
Tajikistan	18.1735197	4.2264803
Tanzania	27.6552221	-3.355222097
Thailand	14.45255213	-5.652552131
Timor-Leste	18.85392103	-8.653921026
Togo	15.10642799	3.193572007
Tonga	15.10805146	1.191948537
Trinidad and Tobago	2.663479112	-1.563479112
Tunisia	7.001424969	2.798575031
Turkiye	8.400199873	-1.900199873
Turkmenistan	10.80652427	0.893475727
Uganda	28.11653972	-4.116539718
Ukraine	7.780710188	0.419289812
United Arab Emirates	2.046267098	-1.246267098
United Kingdom	1.744441399	-0.944441399
United States	2.02574515	-1.02574515
Uruguay	5.231456242	2.068543758
Uzbekistan	12.00074536	11.49925464

Breusch-Pagan test / residuals

LM (Observed value)	50.702
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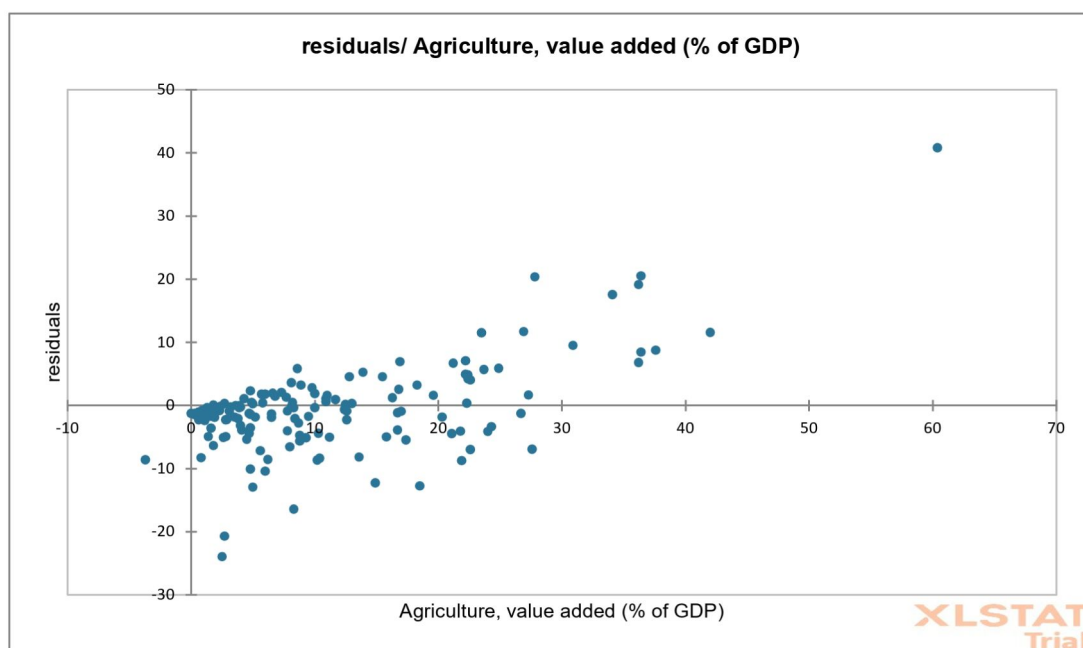
LM (Critical value)	3.841
DF	1
p-value (Two-tailed)	<0.0001
Alpha	0.05

Test interpretation:

H0: Residuals are homoscedastic

Ha: Residuals are heteroscedastic

As the computed p-value is lower than the significance level $\alpha=0.05$, one hypothesis H0, and accept the alternative hypothesis Ha.



White / Squared terms only test / residuals

LM (Observed value)	117.182
LM (Critical value)	5.991
DF	2
p-value (Two-tailed)	<0.0001
alpha	0.05

Test interpretation:

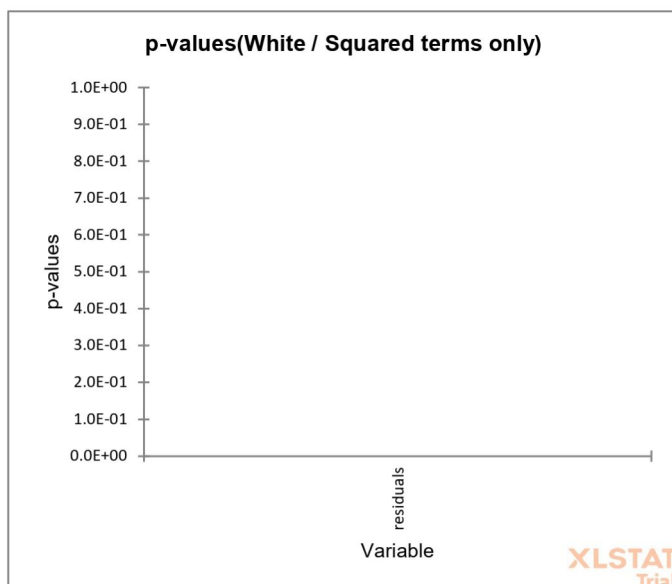
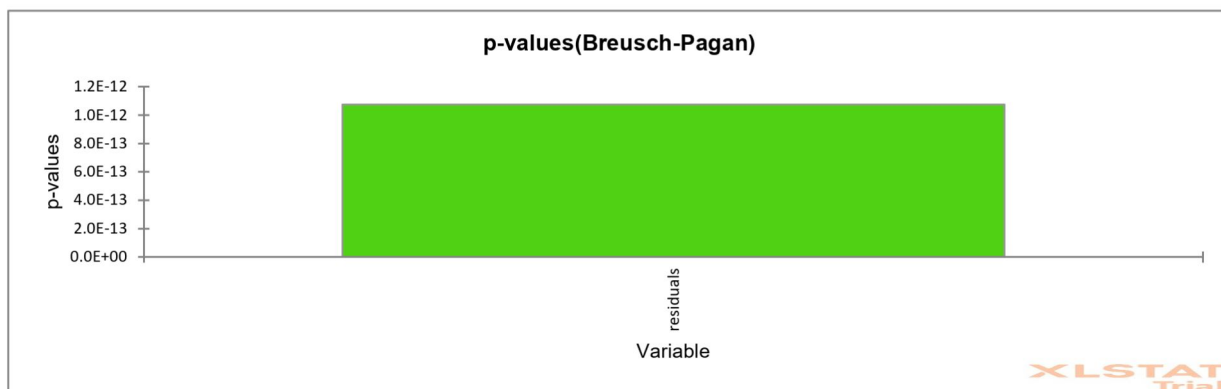
H0: Residuals are homoscedastic

Ha: Residuals are heteroscedastic

As the computed p-value is lower than the significance level $\alpha=0.05$, hypothesis H0, and accept the alternative hypothesis Ha.

Summary:

Variable	Breusch-Pagan	White / Squared terms only
residuals	<0.0001	<0.0001



REMEDIAL MEASURES TO CORRECT HETEROSCEDASTICITY

1. When variance of the population disturbance term is known, then by known variance, we can correct heteroscedasticity.
2. When variance of the population disturbance term is not known.
Case 1 - When it is proportional to the explanatory variable By dividing the explanatory variable, we can correct the Heteroscedasticity.
Case 2 - When it is proportional to the square of explanatory variable By explanatory variable, we can correct the Heteroscedasticity.

TESTING FOR AUTOCORRELATION

The data used here is cross-sectional and no time-series. However, to be sure about it, the Durbin Watson Test has been run on our regression equation to test the presence of autocorrelation. Thus, there are very low chances of the presence of autocorrelation in our model. The Durbin-Watson statistic is approximately 1.85.

It suggests that there is no significant autocorrelation in the residuals.

CONCLUSION

The regression analysis conducted aimed to explore the relationship between gender-specific employment in agriculture and its contribution to GDP across diverse countries. The results indicate a significant positive relationship between male employment in agriculture and agriculture value added (% of GDP), as evidenced by the low p-value associated with the coefficient for male employment. However, the relationship between female employment in agriculture and agriculture value added did not reach statistical significance at the chosen significance level. Despite

this, theoretical considerations and broader datasets suggest that the impact of female employment on agricultural GDP might be substantial, considering factors such as women's roles in agricultural labor, resource access, and productivity contributions.

The findings highlight the importance of considering gender dynamics in agriculture for understanding its overall contribution to economic development. While male participation in agriculture appears to have a significant positive effect on agricultural productivity and GDP, the lack of statistical significance for female employment underscores the need for further investigation. It's possible that the current model may not fully capture the complexities of women's contributions to agriculture, including their roles, access to resources, and the socio-economic context in which they operate.

Moreover, the presence of multicollinearity, indicated by the high correlation between male and female employment in agriculture, suggests that these variables may influence each other and need to be considered together when interpreting the results. Despite this, the absence of significant autocorrelation in the residuals strengthens the reliability of the findings.

In conclusion, while the analysis provides valuable insights into the relationship between gender-specific employment in agriculture and its contribution to GDP, further research is warranted to fully understand the nuanced dynamics at play. Future studies could benefit from incorporating a broader range of variables and a larger dataset to capture the multifaceted nature of gender roles in

agriculture and their implications for economic development. Overall, addressing these gaps in understanding can inform more targeted and effective policy interventions aimed at promoting gender equality and enhancing agricultural productivity globally.

REFERENCES

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